Assessment Part 1: A Report by a Security and Risk Management team

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Introduction **3**
Methodology **4**
Current State **5**
Context & Scope 5
Risk Identification & Risk Analysis 5
Risk Evaluation 5
Risk Treatment & Residual Risk Recalculation 5
Acceptance Rationale 6
Online Business **8**
Context and Scope 8
Proposed Business Changes 8
Implementation of an E-commerce Portal 8
Cloud-Based Inventory Management System (IMS) 8
Customer Relationship Management (CRM) Integration 8
Risk Identification & Analysis 9
Risk Treatment & Residual Risk Recalculation 9
Residual Risk Acceptance Rationale 11
Implementation Priorities 11
Summary **12**
Bibliography **13**
Appendices **14**
Appendix A: Initial Ideas from Molly Fraser 14
Appendix B: Mindmap of Ideas by Deniesha Lloyd 17

Introduction

Pampered Pets is a well-established local business recognised for its high-quality ingredients and personal service. With the introduction of investor Orla O'dour, the business is moving toward full digitalisation and international expansion. This report outlines the risks and mitigations associated with this transition to protect the brand's reputation.

Methodology

This report applies ISO 31000 (2018) and ISO/IEC 27005 (2022) principles. These standards provide a structured approach appropriate for a small business digital transformation. ISO 31000 establishes the overall process of context definition and risk evaluation, while ISO/IEC 27005 focuses on information security risks, such as cloud systems.

A semi-quantitative risk matrix assessed severity, with Likelihood and Impact rated 1–5. Likelihood scores were informed by the NCSC Annual Review, which highlights rising supply chain and small-business targeting in the UK threat landscape. Risk scores are calculated as Likelihood x Impact: 1–7 is 'Low' (Acceptable), 8–14 'Medium' (Tolerable with monitoring), and 15–25 'High' (Unacceptable).

Current State

Context & Scope

The entity is Pampered Pets, a small business with four staff. The scope covers operational and data protection risks within the current on-premise IT environment, including inventory management, sales records, and shared network infrastructure. The objective is to establish a baseline prior to digital transformation.

Risk Identification & Risk Analysis

The business relies on a limited setup, including an older computer for inventory and a single system for sales, creating a single point of failure where hardware issues could halt daily work. Furthermore, the business operates on a shared Wi-Fi network used by both company systems and personal smartphones, increasing the risk of malware or unauthorised access. Human threats are also significant, as staff handle sensitive records without formal cybersecurity training, increasing the likelihood of accidental loss or incorrect data handling.

Risk Evaluation

Risk ID	Threat (Event)	Vulnerability	Likelihood (1-5)	Impact (1-5)	Risk Score (LxI)
C1	Single Point of Failure	Reliance on legacy hardware for inventory	3	4	12 (Medium)
C2	Network Intrusion	Shared Wi-Fi infrastructure	3	4	12 (Medium)
C3	Human Error	Lack of cybersecurity awareness	3	4	12 (Medium)

Risk Treatment & Residual Risk Recalculation

Proposed treatments include weekly encrypted offline backups, network segmentation (Guest Wi-Fi), and NCSC-aligned staff training. Following these controls, the Residual Risk Score becomes 6 (Low). This is accepted by the Business Owner (Risk Sponsor) as no internet-facing systems or online payments exist in the current state.

Risk ID	Initial Score	Chosen Treatment	Proposed Control Measure	Residual Likelihood	Residual Impact	Residual Risk Score
C1: Data Loss	12 (Med)	Modify	Implement weekly encrypted offline backups for Harry's warehouse spreadsheet.	2	3	6 (Low)
C2: Wi-Fi Breach	12 (Med)	Modify	Network segmentation: create a dedicated "Guest Wi-Fi" for staff personal smartphones.	2	3	6 (Low)
C3: Human Error	12 (Med)	Modify	Basic cybersecurity awareness training for all 4 staff members (NCSC, n.d.).	2	3	6 (Low)

Acceptance Rationale

Following these treatments, the Residual Risk is reduced to 6 (Low) across all areas. This is considered an "Acceptable" level for the current business model because:

1. Limited Exposure: The business does not currently store high-value online payment data or operate an e-commerce storefront.
2. Resource Constraints: For a 4-person team, these low-cost manual controls provide sufficient protection until the digitalisation investment is triggered.

Online Business

Context and Scope

Entity: Pampered Pets (Small business, 4 staff).
Scope: This assessment covers operational, information security, and data protection risks within the current on-premise IT environment, including inventory management systems, sales records, shared network infrastructure, and staff handling of business data.
Objective: To identify and evaluate risks affecting business continuity, data integrity, and regulatory compliance in the current state, and to establish a structured baseline prior to digital transformation

Proposed Business Changes

Implementation of an E-commerce Portal

Scalability: Transitions the business from manual, high-friction email orders to an automated platform (e.g., Shopify/WooCommerce), expanding market reach beyond Hashington-on-the-Water.

Growth Potential: Taps into a UK pet e-commerce market that holds a 20% online share (ECDB, n.d.), supporting a projected 50% revenue increase.

Customer Expectations: Meets modern "omnichannel" demands for seamless "Click & Collect" services (Watson, 2021).

Brand Trust: Leverages the "humanisation" trend for premium, local ingredients (Clifford, 2025) and aligns with FEDIAF (2025) nutritional standards to build online credibility.

Cloud-Based Inventory Management System (IMS)

Operational Resilience: Replaces high-risk legacy hardware and manual spreadsheets, a "single point of failure", with a synchronised, real-time cloud environment.
Data Security: Enhances disaster recovery and data resilience, as recommended for small businesses (NCSC, n.d.; Abdi, Bennouri, and Keane, 2024).
Efficiency: Automates in-house preparation and provides the "real-time visibility" necessary to maintain a reliable 10-minute supply chain while increasing order volume (SpendEdge, 2024).

Customer Relationship Management (CRM) Integration

Risk Mitigation: Combats a potential 33% customer churn rate by adopting modern, data-driven loyalty features.
Personalisation: Uses specific data (e.g., pet birthdays) to deliver automated, targeted marketing.

Engagement: Addresses the "pet-as-family" consumer mindset, where personalised advice acts as a primary driver for retention (Multidev Technologies, 2025).
Longevity: Shifts the business model from simple transactions to deep, digital brand loyalty, ensuring long-term competitiveness.

Risk Identification & Analysis

We have identified the top threats associated with your three proposed changes.

Risk ID	Threat (Event)	Vulnerability	Likelihood (1-5)	Impact (1-5)	Risk Score (LxI)
R1	Data Breach (CRM)	Weak employee passwords or lack of MFA on the CRM	4	5	20 (High)
R2	System Downtime	Reliance on a single internet gateway for the cloud IMS	3	4	12 (Medium)
R3	Payment Fraud	Phishing or spoofed checkout pages on the e-commerce site	4	4	16 (High)
R4	Human Error	Accidental deletion of inventory data by staff (Alice, Harry, etc.)	3	3	9 (Medium)
R5	Supply Chain Breach	Vulnerabilities in a 3rd-party Shopify plugin	2	4	8 (Medium)

Risk Treatment & Residual Risk Recalculation

In accordance with ISO 27005, each "Unacceptable" or "Medium" risk identified in the online transition must be addressed through a formal treatment plan. The following table details the treatment options (Modify, Share, or Avoid) and recalculates the Residual Risk, which is the threat level that remains after the security controls are implemented.

Risk ID	Initial Score	Chosen Treatment (ISO 27005)	Proposed Control Measure	Residual Likelihood	Residual Impact	Residual Risk Score
R1: Data Breach	20 (High)	Modify	Mandatory MFA on CRM/Admin accounts and quarterly staff "Human Firewall" training.	2	4	8 (Medium)
R2: Downtime	12 (Med)	Modify	Installation of a 5G failover internet gateway to ensure IMS and portal connectivity.	2	3	6 (Low)
R3: Payment Fraud	16 (High)	Share	Outsourced payment processing to PCI-DSS compliant providers (Stripe/PayPal), transferring liability.	2	2	4 (Low)
R5: Supply Chain	8 (Med)	Modify	Vet all 3rd-party Shopify plugins for security certifications and limit permissions.	1	4	4 (Low)

Residual Risk Acceptance Rationale

The Business Owner, acting as the Risk Sponsor, formally accepts these residual risks. R1 is accepted with continuous monitoring and biannual reviews, as the impact remains high despite technical safeguards.

- Data Breach (R1): While the impact remains significant due to the nature of PII, the Likelihood is significantly lowered by MFA, making it a manageable operational risk.
- Payment Fraud (R3): By "Sharing" this risk with global providers, the business removes the technical burden of securing credit card data, effectively mitigating the threat of a direct financial heist.

Implementation Priorities

To ensure a smooth transition funded by Orla O’dour, the team should prioritise implementation in this order:

1. Immediate (Week 1): Secure the "Digital Identity." Force MFA on all staff devices and smart phones (NCSC, 2024).
2. Short-term (Weeks 2–4): Launch the E-commerce portal with hardened security settings. Transfer Harry’s spreadsheet data into the Cloud IMS with a verified backup.
3. Medium-term (Month 2+): Integrate the CRM. Start with pet birthdays but keep the data entry limited until staff are fully trained on data privacy.

Summary

The risk assessment conducted under the ISO 31000 and ISO 27005 frameworks confirms that Pampered Pets' current manual state is unsustainable. While the "status quo" appears safe, it contains unacceptable "single points of failure", notably Harry’s unbacked-up legacy hardware, which pose a high risk of total data loss.

The transition to a digitalised model is strategically essential to mitigate a projected 33% customer churn and capture the 20% online market share identified by ECDB (n.d.). Although digitalisation introduces high-impact cyber threats such as data breaches (R1) and payment fraud (R3), these are successfully moderated through the proposed risk treatment plan. By implementing Technical Controls (MFA and encrypted cloud backups) and Organisational Controls (staff training), the residual risk is reduced to a "Low" tolerable level.

Therefore, it is recommended that the business proceeds with the investment from Orla O’dour to ensure long-term resilience and achieve the 50% growth target.

Bibliography

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Appendices

Appendix A: Initial Ideas from Molly Fraser

Initial Thoughts
Question 1: Could an online presence grow the business by up to 50%?
Answer: Yes.

- Reasoning: Currently, 90% of sales are local foot traffic. By implementing an e-commerce platform, the "addressable market" shifts from a single suburb to the entire country. High-quality, locally sourced pet food is a massive trend (humanisation of pets).
- Risk: Growth of 50% requires a complete overhaul of Harry's warehouse management and shipping logistics.

Question 2: Could changing to an international supply chain reduce costs by up to 24%?
Answer: Maybe (but proceed with caution).

- Reasoning: Bulk buying ingredients from international markets usually lowers "cost per unit." However, the business's USP (Unique Selling Point) is quality and local sourcing.
- Risk: Switching to international suppliers could damage the brand reputation and introduce supply chain "shocks" (shipping delays, customs), which negates the "10-minute drive" security they currently have.

Question 3: Could the business lose 33% of its existing customers without online features?
Answer: Yes.

- Reasoning: Consumer habits are shifting. If a local competitor offers "click and collect" or a subscription model for heavy bags of dog food, Pampered Pets' customers will prioritise convenience.
- Risk: Loyalty only goes so far; if the friction of buying (driving, parking, manual emailing) is too high, customers will churn.

Risk Management Frameworks
I recommend OCTAVE Allegro or ISO/IEC 27005.

Feature	**Current State**	**Digitalised State**
Primary Threat	Physical theft or hardware failure	Cyber attacks, data breaches, DDoS
Vulnerability	Single point of failure	Unsecured IoT devices and customer data storage

Impact	Localised loss of records	Global reputation allows damage and GDPR fines

Proposed Digitalisation Changes

E-commerce Portal: Moving from manual emails to a Shopify/WooCommerce site.

Inventory Management System (IMS): Replacing Harry's spreadsheet with a cloud-based system that updates in real-time.

CRM (Customer Relationship Management): To track pet birthdays and send automated marketing, boosting loyalty.

However, risks aren't limited to digital tools…the biggest risk is typically the people operating the tools. The employees will need training, which costs time and money, and the biggest risk in any digitalisation is user resistance and lack of technical skill.

Sources

1. Market Growth & Trends (Backing the 50% Growth)

These sources provide the data needed to justify why an online presence could significantly increase revenue.

- UK Pet Food Market Report 2025 (Mintel): This report highlights that "value growth will be driven by trading up" and identifies a massive trend toward "homemade" and "natural" pet foods…directly matching Pampered Pets' USP. [Source: Mintel Store]
- UK Pet Supplies E-Commerce Market 2024-2029 (ECDB): Statistics show that the UK pet e-commerce market reached an online share of 15-20% in 2024. For a business currently at 0% online sales, capturing even a fraction of this national market supports the 50% growth theory. [Source: ECDB]
- European Pet Food Facts and Figures 2025 (FEDIAF): Confirms that 49% of European households own a pet and that the market grew by 9% in a single year (2023-2024), indicating a "blooming" industry ready for digital expansion. [Source: PET Worldwide]

2. Supply Chain Analysis (International vs. Local)

We can use these to argue the "Maybe/No" regarding switching to international suppliers for a 24% cost reduction.

- 7 Key Benefits of Short Food Supply Chains (SpendEdge): This source details how "Short Food Supply Chains" (like Pampered Pets' 10-minute drive) reduce risk, increase quality, and improve consumer trust…factors often lost when moving to international sourcing. [Source: SpendEdge]
- Global Pet Food Volatility & Resilience 2026 (Fastmarkets): Explains that while international sourcing seems cheaper, "localised events can instantly become global

pricing crises" due to tariffs and climate impacts on crops, potentially wiping out that 24% cost saving. [Source: Fastmarkets]
- Sustainable Development in SFSC (MDPI): A critical literature review showing that short supply chains are a "necessary condition" for high-quality food production, which is the core of Pampered Pets' reputation. [Source: MDPI]

3. Cybersecurity & Risk Management (Methodology)
These are the "Gold Standard" sources for your Risk Assessment methodology.

- NCSC Small Business Guide (UK National Cyber Security Centre): The essential guide for small shops. It provides a 5-step framework: Backing up data, Malware protection, Keeping smartphones safe, Password security, and Avoiding phishing. [Source: NCSC/NBCC]
- NCSC Annual Review 2024/2025: Provides the "Threat Landscape" data your report needs, specifically highlighting how "supply chain attacks" and "vulnerabilities in smaller suppliers" are rising. [Source: ISMS.online]
- SANS Institute: Network Security for SMBs: A whitepaper specifically for Small and Mid-sized Businesses (SMBs) that helps prioritise threats when transitioning from a manual to a digital environment. [Source: SANS Institute]

4. Consumer Behavior & Digital Features (Backing the 33% Loss)
These sources back the idea that customers will leave if the shop doesn't modernise.

- 8 Pet Industry Trends for 2025 (Multidev): Reports that 70% of shoppers now demand "omnichannel" features, such as the ability to buy online and pick up in-store (Click & Collect). Without this, Pampered Pets is at risk of losing a third of its base. [Source: Multidev]
- Digital Pet Landscape UK (GlobalPETS): Reveals that the fastest-growing categories are subscription-based pet foods. If Pampered Pets doesn't offer a digital subscription for their premium food, customers may switch to brands that do. [Source: GlobalPETS]
- Loyalty in Pet Retail (DMN): Explains that traditional pet retailers lose customers because their loyalty programs lack the "personalisation or convenience" of digital-first competitors. [Source: DMN.ca]

Appendix B: Mindmap of Ideas by Deniesha Lloyd

